Keywords: video generation • visual reasoning • benchmark • diffusion models

VGI-Bench: Probing Visual Intelligence in Video Generation Models

27 Tasks and 810 Instances Reveal That Even the Strongest Video Generator Fails Half the Time on Visually Grounded Reasoning

By Xuan He et al. (23 authors)

arXiv:2608.19583 • Preprint, August 2026

Video generation models have surpassed human perceptual quality ratings on many production metrics, but can they *reason visually* through their generated frames? VGI-Bench probes this question with 27 tasks spanning physics, spatial logic, counting, and rule-following, each scored by an executable verifier that accepts any feasible solution. The answer is sobering: Seedance 2.0, the strongest evaluated model, achieves 51.0%; denoising trajectory analysis reveals that models commit to visual hypotheses early and almost never self-correct.

AT A GLANCE

Problem: Existing video generation benchmarks reward perceptual plausibility rather than visually grounded reasoning; current models may look good while failing to satisfy physical or logical constraints.

Why it matters: Video generation is increasingly used for simulation, planning, and reasoning augmentation; measuring visual intelligence is essential to evaluate genuine utility.

Method: 27 tasks × 810 instances with a two-level taxonomy; executable verifiers accept any feasible solution (not just ground truth); inputs designed for video model visual priors.

Result: Seedance 2.0 achieves 51.0% Verifier Acceptance Rate (VAR); most models score below 40%; denoising analysis shows no self-correction.

Evidence: Physical constraint tasks are hardest (38%); failure modes: physical collapse (31%), rule violation (28%), object inconsistency (24%).

The Big Picture

Video generation models are no longer merely artists. They are increasingly deployed as simulators for planning, world-model probes, and reasoning-augmented generation — roles that require genuine visual intelligence, not just surface realism.

Recent work has suggested that video generation models may exhibit emergent visual reasoning: given a visual prompt depicting an initial state, can the model generate a video that evolves the scene in a physically and logically correct manner toward a goal? This would indicate that the model has learned not just to hallucinate plausible footage, but to solve visual problems through generation.

VGI-Bench is the first benchmark designed to test this hypothesis systematically, with tasks that have verifiable solutions and inputs matched to the visual priors of contemporary video generators.

Why Existing Approaches Fall Short

Plausibility vs. correctness: Existing benchmarks (VBench, FID/FVD-based metrics) measure perceptual quality — whether generated videos look realistic. A video can be perfectly realistic while showing physically incorrect dynamics or failing to satisfy an explicit constraint. Perceptual metrics cannot measure whether the model solved the visual problem.

Input misalignment: Benchmarks designed for discriminative VLMs use image-caption pairs as input, which do not match the generation-conditioned visual priors of diffusion video models. Inputs must be designed as visual scenes appropriate for the model's training distribution.

Benchmark saturation: Benchmarks with simple visual tasks are rapidly saturated by frontier models, providing no differentiation above a certain capability threshold.

Core Mechanism

VGI-Bench specifies tasks via **hidden structured programs** that define:

- An initial visual scene (rendered as an image prompt)
- A goal specification or constraint set
- An executable verifier that accepts any feasible solution

The verifier is key: it does not compare to a fixed ground-truth video. Instead it takes the final frame (or frame sequence) and checks whether the specification is satisfied. This allows diverse valid solutions and avoids rewarding models that happen to generate the same pixels as a reference.

The **two-level taxonomy** crosses:

Task domain (9): Physics simulation, spatial arrangement, counting, pattern completion, causal reasoning, rule-following, logical deduction, navigation, object tracking

Skill tag (5): Constraint satisfaction, state tracking, multi-step planning, self-correction, multi-object coordination

The 27 tasks are chosen from the 9×5 crossing to maximise domain and skill diversity while maintaining calibrated difficulty (tasks are neither trivial for current models nor completely unsolvable).

Technical Formulation

Let $\mathcal{P}$ be the space of structured specifications and $f : \mathcal{P} \to \mathcal{S}$ a rendering function that maps a specification to a visual scene. A video model G generates a video $v = G(f(p))$ conditioned on the rendered scene.

The verifier $\mathcal{V} : v \times p \to \{0, 1\}$ checks whether video v satisfies specification p. The **Verifier Acceptance Rate (VAR)** over M instances is:

$$\mathrm{VAR}(G) = \frac{1}{M} \sum_{i=1}^{M} \mathcal{V}(G(f(p_i)), p_i)$$

The benchmark evaluates each model with a single generation per instance (no rejection sampling), to measure default generation quality rather than best-of-k performance.

Learning or Inference Procedure

No training occurs within the benchmark. Models generate videos from image prompts using their standard inference pipeline (text conditioning is minimal — only a brief, non-task-specific caption is provided to avoid information leakage).

Denoising trajectory analysis: For diffusion-based models, VGI-Bench provides a methodology for inspecting intermediate denoising steps using a frozen VLM evaluator at regular intervals (every 10% of denoising steps). This reveals whether the model establishes the correct visual structure early or corrects errors later in denoising.

What the Guarantee Says

The verifier acceptance criterion is a sufficient condition for correctness on each task (any feasible solution is accepted), making VAR a lower bound on true task-solving rate. The paper validates verifier accuracy against human ratings: verifier agreement with human binary correctness judgments is 94.2% across a 200-instance subset.

Experimental Findings

- **Seedance 2.0:** 51.0% VAR (strongest model tested)
- **Wan-2.1, Kling-2.0, others:** Below 40% VAR
- **Per-domain:** Physics constraints hardest (Seedance 2.0: 38%); counting/arrangement most accessible (Seedance 2.0: 67%)
- **Per-skill:** Multi-step planning tasks show largest cross-model spread, confirming they are capability-differentiating
- **Denoising trajectories:** Models set visual structure in first ∼20% of denoising steps; later steps add detail without correcting structural errors; self-correction rate: <4% of initially incorrect states across all models

Failure mode breakdown:

- Physical collapse (gravity, collision violations): 31%
- Rule violation (incorrect step in evolving system): 28%
- Object/state inconsistency (identity drift): 24%
- Correct terminal state, incorrect trajectory: 17%

Ablations and Interpretation

Text conditioning: Adding task-specific text prompts (hint at the constraint) improves VAR by +7.4 pp on average, but models remain far from ceiling, suggesting limited text-to-visual reasoning integration.

Best-of-k generation: Best-of-8 using the verifier as an oracle oracle selector improves Seedance 2.0 to 66.2% VAR, showing that partial capability exists but is unreliably expressed in a single generation.

Taxonomy analysis: Skill tag "constraint satisfaction" is the weakest across models (mean VAR 32%); "counting and arrangement" is the strongest (mean VAR 51%). This suggests that spatial counting and arrangement has been inadvertently trained, while hard constraint satisfaction has not.

Reference

Xuan He et al. **VGI-Bench: Probing Visual Intelligence in Video Generation Models.** arXiv:2608.19583, August 2026. https://arxiv.org/abs/2608.19583